

The *ALF* (Algorithms for Lattice Fermions) project release 1

Tutorial

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Exercise 1) Testing against ED

Run the code with the Mz choice of Hubbard Stratonovitch transformation on a four site ring, at $U/t = 4$ and inverse temperature $\beta t = 2$. For this set of parameters, the exact internal energy reads:

$$\langle -t \sum_{\langle i,j \rangle, \sigma} c_{i,\sigma}^\dagger c_{j,\sigma} + U \sum_{i=1}^N n_{i,\uparrow} n_{i,\downarrow} \rangle = -1.47261997t \quad (1)$$

To reproduce this result you will have to carry out a systematic $\Delta\tau t$ extrapolation keeping $\Delta\tau t L_{\text{Trotter}} = 2$ constant. Recall that the formulation of the auxiliary field QMC approach is based on the Trotter decomposition

$$e^{-\Delta\tau(A+B)} = e^{-\Delta\tau A} e^{-\Delta\tau B} + \mathcal{O}(\Delta\tau^2) \quad (2)$$

The overall error produced by this approximation is of the order $\Delta\tau^2$.

All the files you will need to run the code can be found in the directory `Exercise_1/Start`. Beware to define one directory per run so as to avoid conflicts between different parameter sets. Such conflicts will lead to crashes. Here is the parameter file you should use

```
!=====
! Variables for the Hubb program
!-----

&VAR_lattice
L1 = 4           ! Length in direction a_1
L2 = 1           ! Length in direction a_2
Lattice_type = "Square" ! a_1 = (1,0), a_2=(0,1), Norb=1, N_coord=2
!Lattice_type = "Honeycomb" ! a_1 = (1,0), a_2 = (1/2,sqrt(3)/2), Norb=2, N_coord=3
!Model = "Hubbard_SU2" ! Sets Nf=1, N_sun=2. HS field couples to the density
Model = "Hubbard_Mz" ! Sets Nf=2, N_sun=1. HS field couples to the
! z-component of magnetization.
!Model="Hubbard_SU2_Ising" ! Sets Nf_1, N_sun=2 and runs only for the square lattice
! Hubbard model coupled to transverse Ising field

/

&VAR_Hubbard
! Variables for the Hubbard model
ham_T = 1.D0 ! Hopping parameter
ham_chem= 0.D0 ! chemical potential
ham_U = 4.D0 ! Hubbard interaction
Beta = 2.D0 ! inverse temperature
dtau = 0.1D0 ! Thereby Ltrot=Beta/dtau

/

&VAR_QMC
! Variables for the QMC run
Nwrap = 10 ! Stabilization. Green functions will be computed from
! scratch after each time interval Nwrap*Dtau
```

```

NSweep  = 1000      ! Number of sweeps
NBin    = 10        ! Number of bins
Ltau    = 1         ! 1 for calculation of time displaced Green functions;
                   ! 0 otherwise

LOBS_ST = 1         ! Start measurements at time slice LOBS_ST
LOBS_EN = 100       ! End   measurements at time slice LOBS_EN
CPU_MAX = 0.1       ! Code will stop after CPU_MAX hours.
                   ! If not specified, code will stop after Nbin bins.

/
&VAR_errors        ! Variables for analysis programs
n_skip  = 2         ! Number of bins that will be skipped.
N_rebin = 1         ! Rebinning
N_Cov   = 0         ! If set to 1 covariance will be computed
                   ! for unequal time correlation functions.

/

```

After running the program, `$DIR/Prog/Examples.out`, run the analysis programs, `bash analysis.sh`. Note that you have to set the environment variable `$DIR` to the directory where the package is located. This can be done by running `./setenv.sh` in the home directory of the ALF. You will find the internal energy in the file `Ener_scalJ`. This file contains two entries: `OBS: 1` corresponds to the energy and `OBS: 2` to the average sign. If you want to improve the quality of the data, you can rerun the code. Before doing this be sure to move the `confout_*` files to the `confin_*` files. The bash script `out_to_in.sh` will do this for you. Here is the result you should obtain when choosing $\Delta\tau t = 0.05, 0.1, 0.125$.

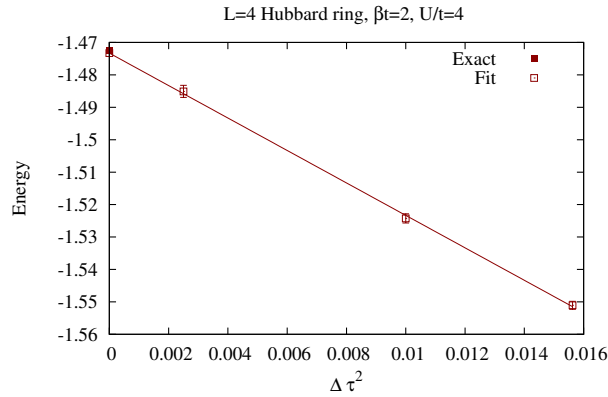


Figure 1: $\Delta\tau t$ extrapolation for the 4-site Hubbard ring.

Exercise 2) Dimensional crossover.

a) Modifying the hopping

Here we will modify the code so as to allow for different hopping matrix elements along the x and y directions of a square lattice. To do so, one merely has to do the following

- Add an extra variable, `Ham_Ty`, in the parameter file in the `VAR_Hubbard` name space
- Declare the variable `Ham_Ty` in the `Hamiltonian_Examples.f90` .
- Read in this variable in `Ham_set` subroutine of the `Hamiltonian_Examples.f90` file.
- Modify the hopping matrix in the subroutine `Ham_Latt` in the `Hamiltonian_Examples.f90` file.

```
DO I = 1, Latt%N
  I1 = Latt%nnlist(I,1,0)
  I2 = Latt%nnlist(I,0,1)
  Op_T(nc,n)%O(I,I1) = cmplx(-Ham_T, 0.d0, kind(0.D0))
  Op_T(nc,n)%O(I1,I) = cmplx(-Ham_T, 0.d0, kind(0.D0))
  !!!!! Modifications for Exercise 2 (a)
  !Op_T(nc,n)%O(I,I2) = cmplx(-Ham_T, 0.d0, kind(0.D0))
  !Op_T(nc,n)%O(I2,I) = cmplx(-Ham_T, 0.d0, kind(0.D0))
  Op_T(nc,n)%O(I,I2) = cmplx(-Ham_Ty, 0.d0, kind(0.D0))
  Op_T(nc,n)%O(I2,I) = cmplx(-Ham_Ty, 0.d0, kind(0.D0))
  !!!!!
  Op_T(nc,n)%O(I ,I) = cmplx(-Ham_chem, 0.d0, kind(0.D0))
ENDDO
```

In the directory `Exercise_2/solutions` we have duplicated the ALF and commented the changes that have to be carried out to the file `Hamiltonian_Examples.f90` in the `Prog` directory.

As an application of this code, one can consider a ladder system, defined by the parameter file:

```
!=====
! Variables for the Hubb program
!-----

&VAR_lattice
L1    = 14
L2    = 2
Lattice_type = "Square"
Model = "Hubbard_Mz" ! Hubbard_SU2, Hubbard_Mz
/

&VAR_Hubbard
ham_T    = 1.0D0
!!!!!! Modifications for Exercise 2
ham_Ty   = 2.0D0
!!!!!!
ham_chem = 0.0D0
ham_U    = 4.0
Beta     = 10.0
dtau     = 0.1
/

&VAR_QMC
Nwrap    = 10
NSweep   = 100
NBin     = 10
Ltau     = 1
LOBS_ST  = 1
```

```

LOBS_EN = 100
CPU_MAX = 0.1
/

&VAR_errors
n_skip = 2
N_rebin = 1
N_Cov = 0
/
! slash terminates namelist statement - DO NOT REMOVE

```

When running the code for the above ladder system and analyzing the spin correlation functions, `SpinZ.eqJR` and `SpinXY.eqJR` one will notice that it is hard to restore the $SU(2)$ spin symmetry and that very long runs are required to obtain the desired equality

$$\langle S_i^z S_j^z \rangle = \langle S_i^y S_j^y \rangle = \langle S_i^x S_j^x \rangle. \quad (3)$$

The structure of the output files `SpinZ.eqJR` and `SpinXY.eqJR` is described in the documentation. For the Mz Hubbard-Stratonovitch transformation it is hence better to consider the improved estimator

$$\langle \vec{S}_i \cdot \vec{S}_j \rangle \quad (4)$$

to compute the spin-spin correlations.

(b) Adding a new observable

Here the aim is to include the new observable equal time observable $\langle \vec{S}_i \cdot \vec{S}_j \rangle$ in the `Hubbard_Mz` code. To achieve this, you will have to carry out the following steps.

- In the subroutine `Alloc_obs` in the `Hamiltonian_example.f90` file you will have to add a new equal time observable with a call to `Call_Obser_Latt_make(Obs_eq(I),Ns,Nt,No,Filename)` with `Ns = Latt%N`; `No = Norb`; `Filename = "SpinT"`, `Nt=1`, `I=5`
- In the subroutine `Obser` you will have to add the Wick decomposition of this observable.

In the program `Hamiltonian_Examples.f90` to be found in the directory `Solutions/Exercise_2/Prog/` we have commented the changes that have to be carried out to add this observable. The new variable takes the name `SpinT` and the results you should obtain are summarized in Fig. 2.

(c) The $SU(2)$ Hubbard Stratonovitch transformation

The $SU(2)$ Hubbard Stratonovitch decomposition, conserves spin rotational symmetry. Run the ladder code with the $SU(2)$ flag in the parameter file switched on (i.e. `Model = Hubbard.SU2`) and compare results.

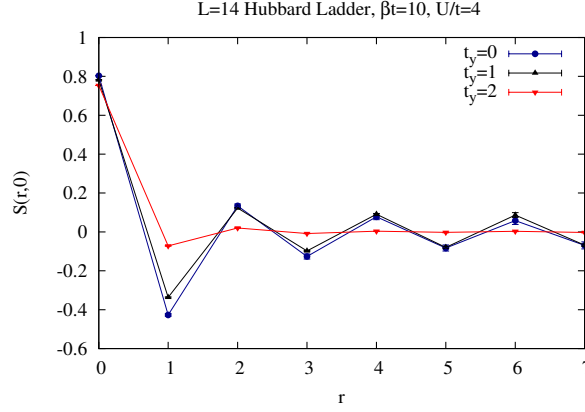


Figure 2: Spin correlation functions along one leg for the Hubbard ladder. As t_y grows the spin gap becomes large enough so as to detect the exponential decay of the spin correlation function on this small lattice size. The underlying physics of odd-even ladder systems is introduced in the article: Elbio Dagotto and T. M. Rice, Surprises on the way from one- to two-dimensional quantum magnets: The ladder materials, Science 271 (1996), no. 5249, 618-623.

Exercise 3) Defining a new model: The one-dimensional t-V model.

(a)

In this section, one we will show what modifications have to be carried out for computing the physics of the one dimensional t-V model of spinless fermions.

$$H = -t \sum_i \left(c_i^\dagger c_{i+a} + c_{i+a}^\dagger c_i \right) - \frac{V}{2} \sum_i \left(c_i^\dagger c_{i+a} + c_{i+a}^\dagger c_i \right)^2 \quad (5)$$

The above form is readily included in the ALF since the interaction is written in terms of a perfect square. Expanding the square yields (up to a constant) the desired model:

$$H = -t \sum_i \left(c_i^\dagger c_{i+a} + c_{i+a}^\dagger c_i \right) + V \sum_i (n_i - 1/2) (n_{i+a} - 1/2) \quad (6)$$

In the directory `Exercise_3/solutions` we have duplicated the ALF and commented the changes that have to be carried out to the file `Hamiltonian_Examples.f90` in the `Prog` directory so as to include the t-V model. Here are the steps to be carried out.

- Add the $t - V$ name space in the parameter file so as to read in the appropriate variables.
- Declare new variables in the `Hamiltonian_Examples.f90` file.
- In the `Ham_set` subroutine of the file `Hamiltonian_Examples.f90` set and read in the parameters for the new model, `t_v`. For this model `NF=1` and `N_SUN=1` since we are working with spinless fermions
- In the `Ham_V` subroutine you will have to add the new interaction. For a given bond at a given time-slice, we need to decouple the interaction:

$$e^{\Delta\tau \frac{V}{2} (c_i^\dagger c_{i+a} + c_{i+a}^\dagger c_i)^2} = \sum_{l=\pm 1, \pm 2} \gamma_l e^{\sqrt{\Delta\tau \frac{V}{2}} \eta_l (c_i^\dagger c_{i+a} + c_{i+a}^\dagger c_i)} = \sum_{l=\pm 1, \pm 2} \gamma_l e^{g \eta_l (c_i^\dagger, c_{i+a}^\dagger) O(c_i, c_{i+a})^T} \quad (7)$$

Here is how this translates in the code.

```
Allocate(Op_V(Latt%N,N_FL))
do nf = 1,N_FL
  do i = 1, N_coord*Ndim
    call Op_make(Op_V(i,nf),2)
```

```

        enddo
    enddo
    Do nc = 1, Latt%N ! Runs over bonds = # of lattice sites in one-dimension.
        I1 = nc
        I2 = Latt%nnlist(I1,1,0)
        Op_V(nc,1)%P(1) = I1
        Op_V(nc,1)%P(2) = I2
        Op_V(nc,1)%O(1,2) = cmplx(1.d0 ,0.d0, kind(0.D0))
        Op_V(nc,1)%O(2,1) = cmplx(1.d0 ,0.d0, kind(0.D0))
        Op_V(nc,1)%g      = SQRT(CMPLX( DTAU*Ham_Vint/2.d0, 0.D0, kind(0.D0)))
        Op_V(nc,1)%alpha  = cmplx(0d0,0.d0, kind(0.D0))
        Op_V(nc,1)%type =2
        Call Op_set( Op_V(nc,1) )
    enddo

```

- Finally you will have to update the **Obser** and **ObserT** routines for the calculation of the equal and time displaced correlations. For the **t_V** model you can essentially use the same observables as for the **Hubbard_SU(2)** model.

You can now run the code for various values of V/t . A Jordan-Wigner transformation will map the **t_V** model onto the XXZ chain:

$$H = J_{xx} \sum_i S_i^x S_{i+a}^x + S_i^y S_{i+a}^y + J_{zz} \sum_i S_i^z S_{i+a}^z \quad (8)$$

with $J_{zz} = V$ and $J_{xx} = 2t$. Hence when $V/t = 2$ we reproduce the Heisenberg model. For $V/t > 2$ the model is in the Ising regime with long-range charge density wave order and is an insulator. In the regime $-2 < V/t < 2$ the model is metallic and corresponds to a Luttinger liquid. Finally, at $V/t < -2$ phase separation between hole rich and electron rich phases occur. Fig. 3 shows typical results.

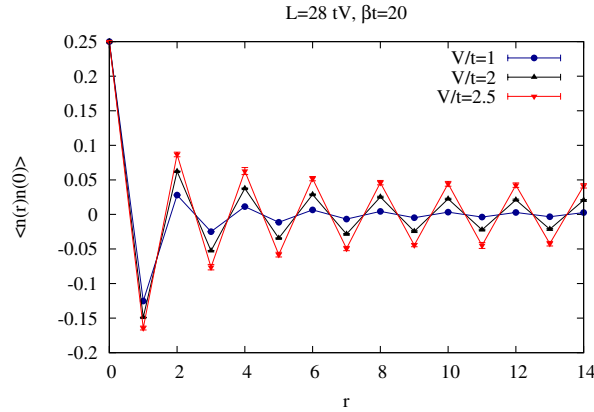


Figure 3: Density-Density correlation functions of the t-V model. In the Luttinger liquid phase, $-2 < V/t < 2$ it is known that the density-density correlations decay as $\langle n(r)n(0) \rangle \propto \cos(\pi r) r^{-(1+K_\rho)}$ with $(1+K_\rho)^{-1} = \frac{1}{2} + \frac{1}{\pi} \arcsin\left(\frac{V}{2|t|}\right)$ (A. Luther and I. Peschel, Calculation of critical exponents in two dimensions from quantum field theory in one dimension, Phys. Rev. B 12 (1975), 3908.) The interested reader can try to reproduce this result.

(b)

How would you use the code to carry out simulations at $V/t < 0$?